Brooke Ballhaus

ballhaus@stanford.edu
linkedin.com/in/brooke-ballhaus | github.com/bballhaus | brookeballhaus.com

AI engineer building systems that separate what should be automated from what is uniquely human, empowering people to use AI and focus on what only they can do.

EXPERIENCE

General Dynamics Information Technology | AI/ML Technical Intern — Falls Church, VA | 06/2024 - Current

- Designed and built PAICUS, a human-in-the-loop AI prototype for CMS's Center for Consumer Information and Insurance Oversight (CCIIO) that helps analysts work through complex healthcare policy
- Proposed, architected, and built a pipeline on AWS to detect fraud, waste, and abuse in 50 million dental claims, using BioBERT embeddings with K-Means and autoencoder anomaly detection, and Claude to write a plain-English explanation of each flagged claim for investigators
- Created and taught "AI: Beyond the Basics," an internal AI training for program managers attended by 60+ people and recorded for company-wide viewing
- Delivered the Day 1 keynote at GDIT's FedCiv Pursuit Palooza 2025, presenting an argument for "Unconstrained Thinking" in AI solution design to Federal Civilian business leaders
- Built and presented LLM tools across 6 projects for the Health Analytics Lighthouse and Luna AI Digital Accelerator, including a multi-model system that automates verification of handwritten healthcare forms; ran SQL spike analysis on federal claims data and delivered findings to the client in Tableau
- Selected from 117 interns to present to the GDIT and General Dynamics Executive Leadership Teams at the August Operations Review; won the end-of-summer award for excellence

NASA Goddard Space Flight Center | Software/Systems Engineering Intern (via Peraton) — Greenbelt, MD | 06/2023 - 05/2024

- Built automated report generation for modem test suites on NASA's Near Space Network (Python backend and GUI), later extended to generate testing procedures

Stanford University | CS107 Teaching Assistant — Stanford, CA | 09/2026 - Current

- Lead weekly lab sections for Stanford's systems programming course, teaching C, memory, and debugging with gdb and Valgrind to 30 students

Stanford Equestrian Team | President — Stanford, CA | 05/2023 - 06/2026

- Led a 40-member team with 35 horses and a $200,000+ annual budget; ran seven home competitions per year and rebuilt the tryout process for ~100 applicants annually

Stanford Society of Women Engineers | Outreach Intern — Stanford, CA | 09/2022 - 06/2026

- Planned six events facilitating 20 mentorships between Stanford and Bay Area high school students interested in STEM

EDUCATION

Stanford University | Master of Science, Computer Science (Artificial Intelligence) — Stanford, CA | Class of 2027

- GPA: 4.0/4.0. Coursework: Agentic AI, NLP with Deep Learning, Deep Reinforcement Learning, Decision Making Under Uncertainty, Statistical Learning, Computer Vision, Ethics of AI, Building Apps with LLMs, Social Computing, Product Management with AI

Stanford University | Bachelor of Science, Computer Science (Artificial Intelligence) — Stanford, CA | Class of 2026

- Activities: Stanford Equestrian Athlete (competed at Zones and Nationals); Ram's Head Theater Actor (three musicals); Alpha Phi Sisterhood Director; Stanford Polo Club Competing Member

PROJECTS

Civitas – AI platform that matches small government contractors with RFPs. Led a 5-person team and wrote the majority of 640+ commits, including scrapers for 62 California procurement sites and semantic matching with Postgres/pgvector

Open Source: crewAI – Two merged PRs to crewAI, a popular multi-agent framework: an offline evaluation runner for agent routing and guardrail decisions (46 tests, mutation-tested) and credential-free AWS Bedrock contract tests; filed the bug they caught (#9095)

Prose.ai (in progress) – Website builder for non-coders where a structured English spec is the source code; each sentence compiles through Claude into its own tested module

Rooted (in progress) – AI decision partner for startup product managers that pulls context from the team's tools, helps weigh options, and keeps a searchable record of what was decided and why

Is Chain-of-Thought Just Rationalization? – Mechanistic audit of chain-of-thought faithfulness in Llama-3-8B (logit lens, activation patching); found the model settles on its answer in the last ~3 of 32 layers with or without chain-of-thought

Saipiens – AI tutor for K-12 that walks students through their own reasoning step by step; 212+ automated tests

SKILLS

Languages & ML: Python, TypeScript, SQL, C; PyTorch, Hugging Face, scikit-learn, TransformerLens, LangChain; Claude/Gemini/Llama APIs; RAG, LLM evals
Systems & Cloud: AWS (Lambda, S3, DynamoDB, RDS, Cognito), PostgreSQL/pgvector, Docker, Next.js, React, Tableau
Certifications: AWS Certified AI Practitioner (2026)